COMP2000 Pattern Implementation Diagram

Traffic simulation - State and Observer patterns

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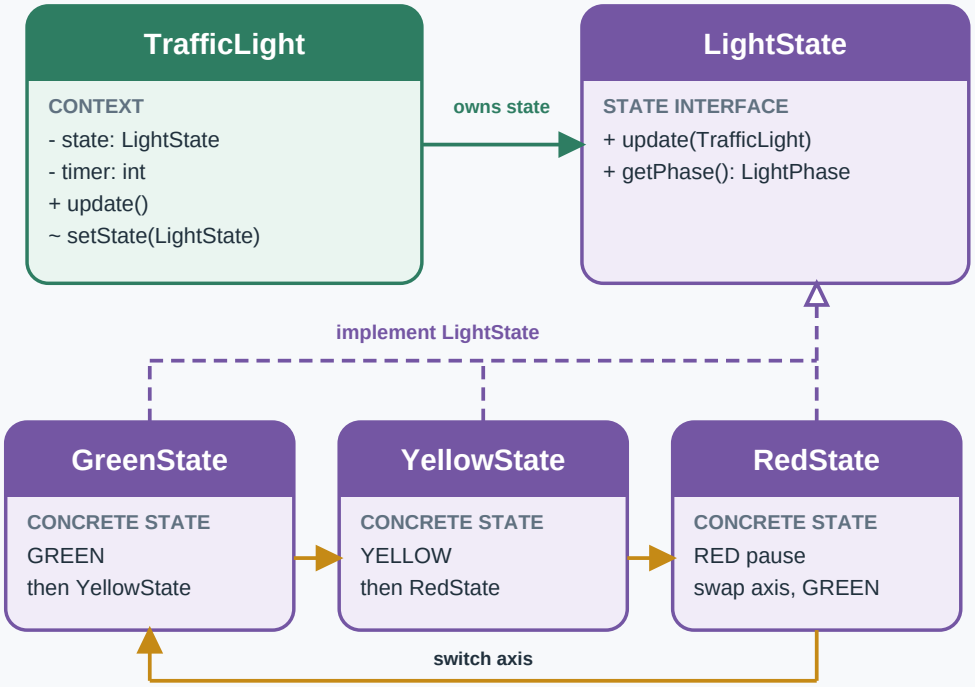
1. State pattern

TrafficLight delegates phase-specific transitions to its current LightState.

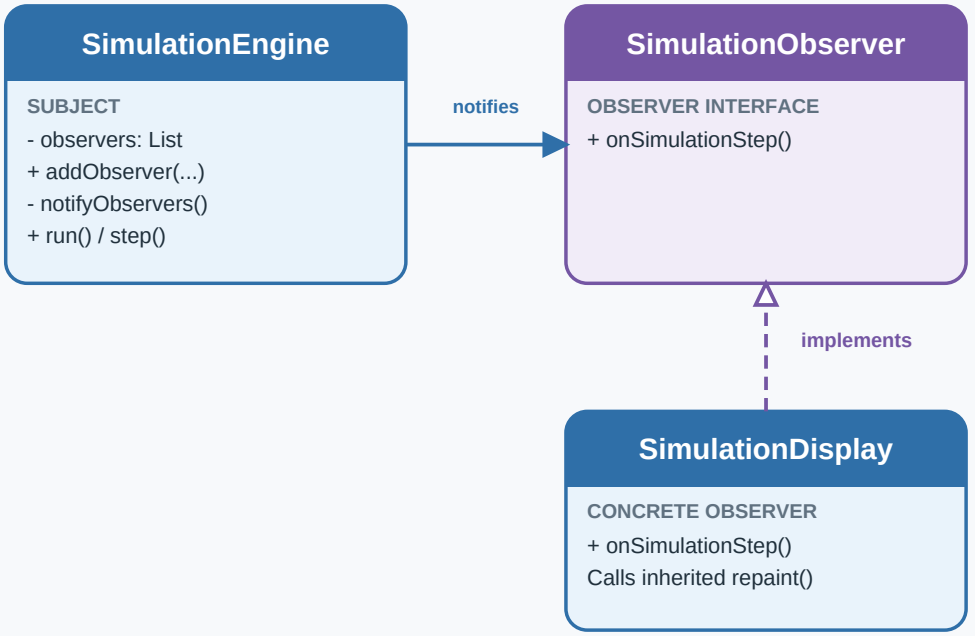


WHY THIS PATTERN

Each phase owns its transition rule. This avoids a growing switch inside TrafficLight. The trade-off is several small classes.

2. Observer pattern

SimulationEngine publishes completed ticks without depending on Swing.



RUNTIME SEQUENCE

1. Engine completes one locked tick.
2. Network lock is released.
3. notifyObservers() calls onSimulationStep().
4. The display requests a repaint. Swing paints under the lock.

Dashed triangle = implements interface Gold arrow = phase transition

Source packages: trafficsim.model.light, trafficsim.engine, trafficsim.view

Main registers the display with engine.addObserver(display) before starting the simulation loop.